

Aadarsh Vimal Singh

Software Engineer | Full-Stack · ML · Cloud Infrastructure

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PROFESSIONAL SUMMARY

B.Tech Computer Science student (ITM Skills University, 2024–28) with production-grade experience spanning ML engineering, cloud infrastructure on AWS EC2, component-driven React frontends, and full-cycle ERP system development. Built and deployed 8 independent projects from architecture to live server — including a cloud-hosted ERP with normalized MySQL schemas, a Scikit-Learn regression pipeline trained on 50K+ records, and a modular React SPA. Writes Python, JavaScript, C++, and Java; administers Linux environments via SSH/SCP/Bash. Motivated by systems thinking and engineering outcomes.

EDUCATION

B.Tech – Computer Science Engineering	ITM Skills University, Kharghar	2024–2028 (Pursuing)
Higher Secondary (HSC)	St. Lawrence High School & Jr. College	Completed 2024
Secondary School Certificate (SSC)	Little Flower High School	Completed 2022

TECHNICAL SKILLS

Languages:	Python, JavaScript (ES6+), TypeScript (Basics), C++, Java, SQL, Bash	Cloud & Infra:	AWS EC2, Ubuntu Server, SSH, SCP, Bash, Security Groups, Cloud Deploy
Frontend:	React.js, HTML5, CSS3, Tailwind CSS, REST API Integration, DOM API	DSA & Systems:	Data Structures, Algorithms, OOP, SOLID Principles, Complexity Analysis
UI/UX:	Figma, Wireframing, Prototyping, Component Design Systems, Canva	Dev Tools:	Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Linux CLI
ML & Data:	Scikit-Learn, Pandas, NumPy, Regression/Classification, EDA, Matplotlib	Testing:	Manual Testing, Unit Test Generation via LLM, Debugging, Code Review
Database:	MySQL, DBMS, Schema Design, Query Optimization, Normalization	Networking:	TCP/IP, OSI Model, Cisco Packet Tracer, HTTP/S Fundamentals
Backend:	Node.js (Learning), REST APIs, JSON, Postman, MVC Architecture	AI Dev Tools:	GitHub Copilot, ChatGPT — rapid prototyping, refactoring & test scaffolding

Soft Skills: Systems Thinking · First-Principles Problem Solving · Technical Communication · Rapid Skill Acquisition · Execution Under Constraints · Collaborative Code Review

PROJECTS

Cloud-Based ERP & Billing System

Python · MySQL · AWS EC2 · SSH · SCP · Bash | Cloud Infrastructure & Backend Engineering

- Architected and deployed a multi-module ERP platform (inventory, invoicing, double-entry accounting) to a live AWS Ubuntu EC2 instance — configuring inbound/outbound security groups, SSH key-pair authentication, and runtime environment variables from a blank server state.
- Designed a fully normalized MySQL relational schema across products, invoices, and ledger tables with enforced referential integrity; structured for horizontal extensibility (HR, CRM, payroll modules) without schema rewrites.

AWS EC2 Server Provisioning & Deployment Pipeline

AWS EC2 · Ubuntu 22.04 · SSH · SCP · Bash · apt | DevOps & Infrastructure

- Provisioned a production-grade cloud server from a blank EC2 Ubuntu instance — establishing SSH key-pair auth, configuring security group firewall rules, and bootstrapping runtime dependencies via apt and Bash init scripts.
- Engineered a repeatable artifact deployment workflow using SCP for binary transfers and shell scripts for environment setup, reducing deployment steps to a single-command execution.

React & Tailwind CSS Single-Page Application

React.js · Tailwind CSS · JavaScript ES6+ · Hooks API | Frontend Engineering

- Engineered a component-driven SPA with a dynamic book list manager and a real-time quote/countdown timer — using useState, useEffect, and setInterval for state lifecycle and timed DOM updates.
- Eliminated custom stylesheet complexity by adopting Tailwind's utility-first system; achieved full mobile-to-desktop responsiveness through a reusable, decoupled component architecture.

Uber Fare Prediction — ML Regression Pipeline

Python · Scikit-Learn · Pandas · NumPy · Matplotlib | Machine Learning Engineering

- Constructed an end-to-end supervised learning pipeline on 50K+ NYC trip records — covering EDA, null imputation, outlier removal, and StandardScaler normalization — then benchmarked Linear Regression, Ridge, and Decision Tree regressors on RMSE and R² to select the optimal model.
- Quantified feature importance through correlation analysis and residual plots, surfacing trip distance and hour-of-day as dominant fare predictors — insights directly applicable to pricing model design.

HACKATHONS & COMMUNITY

- Mumbai Hacks 2024** — Participated in Mumbai's largest student hackathon; executed rapid full-stack prototyping under time constraints, presented live demo to technical judges.
- Mumbai Tech Week 2024** — Attended industry sessions on distributed systems, cloud-native architecture, and AI/ML deployment; engaged directly with engineering leads from top product companies.

CERTIFICATIONS & COURSEWORK

Stock Market & Crypto Trading — Acmegrade · Jan–Feb 2025 | Portfolio construction, IPO analysis, derivatives fundamentals; completed during Semester 1 of B.Tech.
Core Coursework: Data Structures & Algorithms · OOP & Design Patterns · DBMS · Operating Systems · Computer Networks · Software Engineering · Web Technologies · Linear Algebra & Probability

LANGUAGES & INTERESTS

Languages: English (Professional Proficiency) · Hindi (Native) · Marathi (Conversational) | **Interests:** Competitive Programming · Open Source · Distributed Systems · ML/AI Research · Cloud Architecture · Quantitative Finance · Strength Training · Hackathons